

Solution to Exercise 06 — Analysis on Manifolds, 80416

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Question 1

Subquestion 1

Let $M \subseteq \mathbb{R}^d$ be a C^r manifold of dimension m , let $p \in M$ and let $\varphi_i : U_i \rightarrow M$ for $i \in \{1, 2\}$ be two local parameterizations around p . We will show that $\text{Im}(D\varphi_1)_{\varphi_1^{-1}(p)} = \text{Im}(D\varphi_2)_{\varphi_2^{-1}(p)}$.

Proof: Let

$$u_1 = \varphi_1^{-1}(p) \quad u_2 = \varphi_2^{-1}(p)$$

Since φ_1, φ_2 are local parameterizations around p the transition map $\tau = \varphi_2^{-1} \circ \varphi_1$ is well defined and of class C^r on a neighborhood of u_1 and $\varphi_1 = \varphi_2 \circ \tau$.

From the chain rule at u_1 is get that

$$(D\varphi_1)|_{u_1} = (D\varphi_2)|_{u_2} \circ (D\tau)|_{u_1}$$

So for each $v \in \mathbb{R}^m$ we have

$$(D\varphi_1)|_{u_1}(v) = (D\varphi_2)|_{u_2}((D\tau)|_{u_1}(v))$$

Hence every vector in $\text{Im}((D\varphi_1)|_{u_1})$ is in $\text{Im}((D\varphi_2)|_{u_2})$ which means $\text{Im}((D\varphi_1)|_{u_1}) \subseteq \text{Im}((D\varphi_2)|_{u_2})$.

If we reverse the roles of φ_1 and φ_2 we get the $\supseteq$ hence $\text{Im}(D\varphi_1)_{\varphi_1^{-1}(p)} = \text{Im}(D\varphi_2)_{\varphi_2^{-1}(p)}$. $\square$

Subquestion 2

We will conclude from section a that if $M, N \subseteq \mathbb{R}^d$ are two manifolds of dimensions m and n respectively then M intersect N transversally at $p \in M \cap N$ if and only if there exist two local parameterizations around $p \in M \cap N$, $\varphi : U \rightarrow \mathbb{R}^d, \psi : V \rightarrow \mathbb{R}^d$ in M and N respectively such that $\text{Im}(D\varphi)_{\varphi^{-1}(p)} + \text{Im}(D\psi)_{\psi^{-1}(p)} = \mathbb{R}^d$.

Proof: $\Rightarrow$ Assume that M intersect N transversally at $p \in M \cap N$.

This direction is purly from the definition we gave in the recitation: we defined as the following Let $M, N \subseteq \mathbb{R}^d$ be two manifold of dimensions m and n respectively. We say that M intersects N transversally at $p \in M \cap N$ if every two local parameterizations around p , $\varphi : U \rightarrow \mathbb{R}^d, \psi : V \rightarrow \mathbb{R}^d$ where $U \subseteq \mathbb{R}^m, V \subseteq \mathbb{R}^n$ are open sets, satisfy

$$\text{Im}(D\varphi)_{\varphi^{-1}(p)} + \text{Im}(D\psi)_{\psi^{-1}(p)} = \mathbb{R}^d$$

And we said that M intersects N transversally and we denote $M \pitchfork N$ if they intersect transversally at every $p \in M \cap N$.

So this direction is purly from definition.

$\Leftarrow$ Assume that there exist local parametrizations $\varphi : U \rightarrow \mathbb{R}^d, \psi : V \rightarrow \mathbb{R}^d$ such that

$$\text{Im}((D\varphi)|_{\varphi^{-1}(p)}) + \text{Im}((D\psi)|_{\psi^{-1}(p)}) = \mathbb{R}^d$$

From the previous section, we proved that the image of the differential of a local parametrization at p is strictly independent of the parametrization chosen. Therefore, these images uniquely define the tangent spaces

$$T_p M = \text{Im}(D\varphi)_{\varphi^{-1}(p)} \quad \text{and} \quad T_p N = \text{Im}(D\psi)_{\psi^{-1}(p)}$$

So from our assumption we get

$$T_p M + T_p N = \mathbb{R}^d$$

This is exactly the definition of M and N intersecting transversally at p . $\square$

Subquestion 1

For every v in $\mathbb{R}^2$ we consider the set $L_v = \{p + tv : t \in \mathbb{R}\}$.

Proof: Let $p = (x_0, f(x_0)) \in \Gamma(f)$ and let $v = (v_1, v_2) \in \mathbb{R}^2$. If $v = (0, 0)$ then $L_v = \{p\}$ which is a single point and trivially a 0-dimensional manifold.

Otherwise, assume $v \neq (0, 0)$ so we can define a global parametrization $\varphi : \mathbb{R} \rightarrow \mathbb{R}^2$ by $\varphi(t) = p + tv$.

We need to show that this is a local parametrization so we need to show that the domain is open, verify the image, continuity and smoothness and show that it has full rank differential.

Indeed, the domain is $\mathbb{R}$ which is open and $Im(\varphi) = L_v$ and φ is an affine transformation which is C^∞ .

We have that φ is a continuous bijection onto L_v and its inverse is (up to a scalar) the dot product with v which is continuous thus φ is a homeomorphism onto its image.

Last, we have $D\varphi_t = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and since $v \neq 0$ this has full rank hence the differential is injective.

So, L_v is indeed a 1-dimensional C^∞ manifold for $v \neq 0$.

We need to find all L_v such that L_v intersects $\Gamma(f)$ transversally at p .

The graph $\Gamma(f)$ has a global parametrization $\alpha(x) = (x, f(x))$. Its differential at x_0 is the column vector $D\alpha_{x_0} = \begin{pmatrix} 1 \\ f'(x_0) \end{pmatrix}$. Hence

$$T_p\Gamma(f) = \text{Span}\{(1, f'(x_0))\}$$

If $v = 0$, then L_v is a 0-manifold and $T_p L_0 = \{0\}$. The sum of the tangent spaces would just be $T_p \Gamma(f)$, which is 1-dimensional and cannot span $\mathbb{R}^2$. Thus, we must have $v \neq 0$.

If $v \neq 0$, using the parametrization $\varphi(t) = p + tv$, the differential at $t = 0$ is exactly the direction vector v . Thus

$$T_p L_v = \text{Span}\{v\} = \text{Span}\{(v_1, v_2)\}$$

For the sum of these two 1-dimensional subspaces to be the entire 2-dimensional space $\mathbb{R}^2$, their spanning vectors must be linearly independent. Two vectors in $\mathbb{R}^2$ are linearly independent if and only if the determinant of the matrix they form is non-zero

$$\det \begin{pmatrix} 1 & v_1 \\ f'(x_0) & v_2 \end{pmatrix} \neq 0 \iff 1 \cdot v_2 - v_1 \cdot f'(x_0) \neq 0 \iff v_2 \neq v_1 f'(x_0)$$

Therefore, L_v and $\Gamma(f)$ intersect transversally at p if and only if $v \neq 0$ and v is not a scalar multiple of $(1, f'(x_0))$.

Subquestion 2

Let $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$ and for $t \in R$ let $H_t = \{x \in \mathbb{R}^{n+1} : x_{n+1} = t\}$. We will prove that H_t is a manifold and find all $t \in R$ such that $H_t \cap S^n$.

Proof: H_t is a manifold because if we look at $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ defined by $f(x) = x_{n+1} - t$ we get a C^∞ function and its differential is the constant $1 \times (n+1)$ matrix $Df_x = (0 \dots 0 \ 1)$ with a full rank at all points $x \in \mathbb{R}^{n+1}$.

By the definition of manifold as the null set of a system of equations we saw in the recitation, the pre-image $f^{-1}(0) = H_t$ is a smooth manifold of dimension $(n + 1) - 1 = n$.

By definition, H_t and S^n intersect transversally if for every $p \in H_t \cap S^n$, their tangent spaces span the ambient space

$$T_p H_t + T_p S^n = \mathbb{R}^{n+1}$$

[illegible]

Question 3

Subquestion 1

Let $M, N \subseteq \mathbb{R}^d$ be C^r manifolds of dimensions m and n respectively. We will prove that $M \times N \subseteq \mathbb{R}^d \times \mathbb{R}^d = \mathbb{R}^{2d}$ is a C^r manifold of dimension $m + n$.

Proof: Let $(p, q) \in M \times N$.

Because M is a C^r manifold of dimension m in $\mathbb{R}^d$, there exists an open set $U \subseteq \mathbb{R}^m$, an open neighborhood $W_1 \subseteq \mathbb{R}^d$ of p , and a C^r local parametrization $\varphi : U \rightarrow M \cap W_1$ such that the differential $D\varphi_x$ is injective for all $x \in U$.

Similarly, because N is a C^r manifold of dimension n in $\mathbb{R}^d$, there exists an open set $V \subseteq \mathbb{R}^n$, an open neighborhood $W_2 \subseteq \mathbb{R}^d$ of q , and a C^r local parametrization $\psi : V \rightarrow N \cap W_2$ such that the differential $D\psi_y$ is injective for all $y \in V$.

We look at the function $\Phi : U \times V \rightarrow \mathbb{R}^{2d}$ defined by

$$\Phi(x, y) = (\varphi(x), \psi(y))$$

We need to verify that Φ is a valid local parametrization for $M \times N$ around (p, q) so we need to show that the domain is open, verify the image, continuity and smoothness and show that it has full rank differential.

We know that $U \times V$ is open set in $\mathbb{R}^{m+n}$ since product of open sets is open.

The image of Φ is $\varphi(U) \times \psi(V) = (M \cap W_1) \times (N \cap W_2) = (M \times N) \cap (W_1 \times W_2)$ and since $W_1 \times W_2$ is open set in $\mathbb{R}^{2d}$ that contains (p, q) Φ maps onto an open neighborhood of (p, q) in the subspace topology of $M \times N$. Furthermore, because the Cartesian product of two homeomorphisms is a homeomorphism, Φ is a homeomorphism onto its image and each of the coordinates of Φ is of class C^r so this is also smooth.

We need to show that it has $D\Phi$ is of full rank, we have

$$D\Phi_{(x,y)} = \begin{pmatrix} D\varphi_x & 0 \\ 0 & D\psi_y \end{pmatrix}$$

But since $D\varphi_x$ has full rank of m and $D\psi_y$ has full rank of n the block matrix $D\Phi_{(x,y)}$ has full column rank $m + n$. Thus, the differential is injective everywhere in $U \times V$.

We constructed a local parametrization around any (p, q) hence $M \times N$ is a C^r manifold of dimension $m + n$. □

Subquestion 2

Denote $\Delta = \{(x, y) \in \mathbb{R}^d \times \mathbb{R}^d : x = y\}$. We will prove that $\Delta \subseteq \mathbb{R}^{2d}$ is a manifold of dimension d .

Proof: We can prove this immediately by observing that the diagonal Δ is simply the graph of a smooth function.

Define the identity function $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ by $f(x) = x$, the domain of f is $\mathbb{R}^d$ (which is open), and f is a smooth C^r function.

By definition, the graph of f is:

$$\Delta = \Gamma(f) = \{(x, f(x)) : x \in \mathbb{R}^d\} = \{(x, x) : x \in \mathbb{R}^d\}$$

We saw at exercise 5 that the graph of any C^r function $f : U \subset \mathbb{R}^d \rightarrow \mathbb{R}^k$ is a C^r manifold of dimension d . Therefore, Δ is a C^r manifold of dimension d . □

Subquestion 3

We will prove that $M \pitchfork N$ if and only if $M \times N \pitchfork \Delta$.

Proof: **TODOOOOOOOOOOOOOOOOOOO.** □